INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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MARKS											

Homework - 1

3) ⇒ A collection of vectors $\{|\psi_1\rangle, ---, |\psi_l\rangle\}$ is linearly independent if following is true: $\sum_i \alpha_i |\psi_i\rangle = 0$ only when $\alpha_1 = \alpha_2 = ---- = \alpha_l = 0$.

⇒ To prove: If $\{|\psi_i\rangle, --, |\psi_l\rangle\}$. is orthonormal then it is linearly independent.

⇒ $\langle \psi_i | \psi_j \rangle = 0$ if $i \neq j$

$\langle \psi_i | \psi_j \rangle = 1$ if $i = j$

⇒ Take α_i such that $\sum_i \alpha_i |\psi_i\rangle = 0$

⇒ For each $j = 1$ to l,

take $|\psi_j\rangle$ and do apply inner product

$$\langle \psi_j | \left(\sum_i \alpha_i |\psi_i\rangle \right) = (\langle \psi_j |) \cdot 0 = 0$$

$$\sum_i \alpha_i \langle \psi_j | \psi_i \rangle = 0$$

$$0 + 0 + --- + \alpha_j \underbrace{\langle \psi_j | \psi_j \rangle}_{1} + 0 + --- + 0 = 0$$

$$\alpha_j = 0$$

$\Rightarrow \alpha_1 = \alpha_2 = \text{------} = \alpha_l = 0$

$\Rightarrow$ Any orthonormal collection of vectors is linearly independent.

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4) $\Rightarrow$ Let A and B anti-commute. $AB = -BA$.

(a) $\Rightarrow Tr(AB) = Tr(-BA) = -Tr(BA)$

We know that $Tr(AB) = Tr(BA)$ if A and B are well defined.

$\Rightarrow Tr(AB) = -Tr(AB) \Rightarrow \boxed{Tr(AB) = 0}$

(b) $\Rightarrow$ Given A is invertible

~~$\Rightarrow Tr(A^{-1}AB) = Tr(-A^{-1}BA)$~~

$\Rightarrow AB = -BA \Rightarrow A^{-1}AB = -A^{-1}BA$

$\Rightarrow B = -A^{-1}BA \Rightarrow Tr(B) = Tr(-A^{-1}BA)$

$\Rightarrow Tr(B) = -Tr(A^{-1}BA) = -Tr(BAA^{-1}) = -Tr(B)$

(using cyclic property of trace)

$\Rightarrow \boxed{Tr(B) = 0}$

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5)

(a) $\Rightarrow$ Given $\{|\psi_1\rangle, \text{--}, |\psi_n\rangle\}$ is orthonormal set of vectors in C^n. $\longrightarrow$ This forms basis

$\Rightarrow$ To prove: $\forall\ |\phi\rangle \in C^n$, $|\phi\rangle = \sum_{i=1}^{n} |\psi_i\rangle\langle\psi_i|\phi\rangle$

which is a linear combination of $\{|\psi_i\rangle\}$

$\Rightarrow$ Let $|\psi\rangle = |\phi\rangle - \sum_{i=1}^{n} |\psi_i\rangle\langle\psi_i|\phi\rangle$

$\Rightarrow$ For all $j = 1$ to n,

$\langle\psi_j|\psi\rangle = \langle\psi_j||\psi\rangle$

$= \langle\psi_j|\left(|\phi\rangle - \sum_{i=1}^{n} |\psi_i\rangle\langle\psi_i||\phi\rangle\right)$

$$= \langle \psi_j | \phi \rangle - \sum_{i=1}^{n} \langle \psi_j | \psi_i \rangle \langle \psi_i | \phi \rangle$$

$$= \langle \psi_j | \phi \rangle - \sum_{i=1}^{n} [0 + 0 + \cdots + \langle \psi_j | \psi_j \rangle \langle \psi_j | \phi \rangle + 0 + \cdots + 0]$$

[$\because |\psi_i\rangle, |\psi_j\rangle$ belong to orthonormal basis so $\langle \psi_i | \psi_i \rangle = 1$ and $\langle \psi_i | \psi_j \rangle = 1$ for $i \neq j$]

$$= \langle \psi_j | \phi \rangle - \langle \psi_j | \phi \rangle = 0$$

$\Rightarrow \langle \psi_j | \psi \rangle = 0 \quad \forall j = 1$ to n. As $\{|\psi_i\rangle\}_{i=1}^{n}$ forms a basis of $\mathbb{C}^n$, $|\psi\rangle = 0$.

$\Rightarrow$ Hence, $\forall |\phi\rangle \in \mathbb{C}^n$, $|\phi\rangle = \sum_{i=1}^{n} |\psi_i\rangle \langle \psi_i | \phi \rangle$

(b) $\Rightarrow$ Given $\{|\psi_i\rangle\}_{i=1}^{n}$ is any orthonormal set of vectors in $\mathbb{C}^n$ i.e $\{|\psi_1\rangle, \ldots, |\psi_n\rangle\}$ is any orthonormal basis of $\mathbb{C}^n$.

$\Rightarrow$ <u>To prove:</u> $\mathrm{Tr}(A) = \sum_{i=1}^{n} \langle \psi_i | A | \psi_i \rangle$

$\Rightarrow$ Let $\{|e_i\rangle\}_{i=1}^{n}$ is standard basis of $\mathbb{C}^n$

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} A_{ii} = \sum_{i=1}^{n} |e_i\rangle^{+} A |e_i\rangle = \sum_{i=1}^{n} \langle e_i | A | e_j \rangle$$

$\Rightarrow$ We know $\{|\psi_i\rangle\}_{i=1}^{n}$ is a basis.

$$\Rightarrow |\psi_k\rangle = \sum_{i=1}^{n} U_{ik} |e_i\rangle = \sum_{i=1}^{n} \langle e_i | \psi_k \rangle |e_i\rangle$$

where $U_{ik} = \langle e_i | \psi_k \rangle$

$\Rightarrow U = (U_{ik})_{i=1 \text{ to } n,\ k = 1 \text{ to } n}$

$\Rightarrow$ We know that $\langle \psi_k | \psi_l \rangle = \delta_{kl}$

$$\Rightarrow \langle \psi_k | \psi_l \rangle = \left(\sum_i U_{ik}^* \langle e_i|\right)\left(\sum_j U_{jl} |e_j\rangle\right)$$

$$= \sum_{i,j} U_{ik}^* U_{jl} \langle e_i | e_j \rangle$$

$$= \sum_{i,j} U_{ik}^* U_{jl} \delta_{ij} \; \cancel{= \delta_{kl}} = \sum_i U_{ik}^* U_{il}$$

$$\Rightarrow \sum_i U_{ik}^* U_{il} = \delta_{kl} \Rightarrow \boxed{U^\dagger U = I}$$

$\Rightarrow$ U is unitary matrix

$\Rightarrow$ By change of basis of A from standard basis to ψ - basis,

$$\tilde{A}_{kl} = \langle \psi_k | A | \psi_l \rangle$$

$$= \left(\sum_i U_{ik}^* \langle e_i|\right) A \left(\sum_j U_{jl} |e_j\rangle\right)$$

$$= \sum_{i,j} U_{ik}^* \underbrace{(\langle e_i | A | e_j \rangle)}_{A_{ij}} U_{jl}$$

$$= \sum_{i,j} U_{ik}^* A_{ij} U_{jl}$$

$$\Rightarrow \sum_{k=1}^n \tilde{A}_{kk} = \sum_{k=1}^n \sum_{i,j} U_{ik}^* A_{ij} U_{jk}$$

$$= \sum_{i,j} A_{ij} \underbrace{\left(\sum_{k=1}^n U_{ik}^* U_{jk}\right)}_{U \text{ is unitary}} = \sum_{i,j} A_{ij} \delta_{ij}$$

$$\Rightarrow \sum_{k=1}^n \tilde{A}_{kk} = \sum_{i=1}^n A_{ii} = Tr(A)$$

$$\Rightarrow \boxed{Tr(A) = \sum_{i=1}^n \langle \psi_i | A | \psi_i \rangle}$$

7. $\Rightarrow$ A is a square matrix.

(a) $\Rightarrow$ Let $B = \frac{1}{2}A + \frac{1}{2}A^{+}$

$B^{+} = (\frac{1}{2}A + \frac{1}{2}A^{+})^{+} = (\frac{1}{2}A)^{+} + (\frac{1}{2}A^{+})^{+} = \frac{1}{2}A + \frac{1}{2}A^{+}$

$\boxed{B^{+} = B}$ / B is hermitian

$\Rightarrow$ Let $C = \frac{1}{2}A - \frac{1}{2}A^{+}$

$C^{+} = (\frac{1}{2}A + \frac{1}{2}A^{+})^{+} = \frac{1}{2}A^{+} - \frac{1}{2}A = -C$

$\boxed{C^{+} = -C}$ C is skew-hermitian.

(b) $\Rightarrow A = \frac{1}{2}(2A) = \frac{1}{2}(A + A^{+}) + \frac{1}{2}(A - A^{+})$

$\Rightarrow$ A is sum of a Hermitian operator $\frac{1}{2}(A + A^{+})$ and a skew-Hermitian operator $\frac{1}{2}(A - A^{+})$

(c) $\Rightarrow$ Let $H = \frac{1}{2}(A + A^{+})$ and $S = \frac{1}{2}(A - A^{+})$

$\Rightarrow A = H + S$

$\Rightarrow$ Let $B = H$, $\beta = 1$. We know $H^{+} = H$.

$\Rightarrow$ Let $C = iS$

$C^{+} = (iS)^{+} = -iS^{+} = (-i)(-S)$ $[\because S^{+} = -S]$

$= iS = C$.

C is Hermitian. $\longrightarrow S = \frac{C}{i} = -iC$

$\Rightarrow A = H + S = B - iC$, B and C are Hermitian.

$\Rightarrow$ ~~The~~ $\beta = 1$, $\gamma = -i$; $\beta \in \mathbb{C}, \gamma \in \mathbb{C}$.

$\Rightarrow \exists$ hermitian operators B and C, scalars $\beta, \gamma \in C$ such that $A = \beta B + \gamma C$. Every operator is linear combination of Hermitian operators

9. $\Rightarrow$ Positive matrix = PSD matrix

$\Rightarrow$ $n \times n$ operator A is positive when A is hermitian and $\langle\psi|A|\psi\rangle \geq 0 \quad \forall\ |\psi\rangle \in \mathbb{C}^n$

$\Rightarrow$ let ρ be a density operator.

$$\Rightarrow \rho = \begin{bmatrix} p_0 & & & 0 \\ & p_1 & & \\ & & \ddots & \\ 0 & & & p_{n-1} \end{bmatrix}_{n\times n} \qquad p_i \in \mathbb{R},\ 0 \leq p_i \leq 1,\ \sum_{i=0}^{n-1} p_i = 1 = Tr(\rho)$$

$$\Rightarrow \rho^{\dagger} = \begin{bmatrix} \overline{p_0} & & & 0 \\ & \overline{p_1} & & \\ & & \ddots & \\ 0 & & & \overline{p_{n-1}} \end{bmatrix} = \begin{bmatrix} p_0 & & \\ & p_1 & \\ & & \ddots \\ & & & p_{n-1} \end{bmatrix} = \rho$$

ρ is Hermitian

$\Rightarrow$ Take any arbitrary $|\psi\rangle = \begin{bmatrix} \psi_0 \\ \vdots \\ \psi_{n-1} \end{bmatrix} \in \mathbb{C}^n$

$$\Rightarrow \langle\psi|\rho|\psi\rangle = \cancel{[\overline{\psi_0}\ \overline{\psi_1} \cdots \overline{\psi_{n-1}}]}$$

$$= [\overline{\psi_0}\ \overline{\psi_1} \cdots \overline{\psi_{n-1}}] \begin{bmatrix} p_0 & & & 0 \\ & p_1 & & \\ & & \ddots & \\ 0 & & & p_{n-1} \end{bmatrix} \begin{bmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{n-1} \end{bmatrix}$$

$$= [p_0\overline{\psi_0}\quad p_1\overline{\psi_1} \cdots p_{n-1}\overline{\psi_{n-1}}] \begin{bmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{n-1} \end{bmatrix}$$

$$= p_0\psi_0\overline{\psi_0} + p_1\psi_1\overline{\psi_1} + \cdots + p_{n-1}\psi_{n-1}\overline{\psi_{n-1}}$$

$$= \sum_{i=0}^{n-1} p_i |\psi_i|^2 \geq 0 \Rightarrow \langle\psi|\rho|\psi\rangle \geq 0 \quad \forall\ |\psi\rangle \in \mathbb{C}^n$$

[$\because p_i \geq 0,\ |\psi_i|^2 \geq 0 \quad \forall\ i = 0$ to $n-1$]

~~$\Rightarrow \rho$~~

$\Rightarrow$ Density operator corresponding to any classical state (pure or mixed) is a positive operator

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10.

(a) $\Rightarrow$ Alphabet $= \mathcal{X} = \{000, 001, 010, 011, 100, 101, 110, 111\}$

$\Rightarrow P(X = s) \propto \#$ 1's in 's' string

$\Rightarrow$ let $P(X = 000) = 0$

$P(X = 001) = P(X = 010) = P(X = 100) = k$

$P(X = 011) = P(X = 101) = P(X = 110) = 2k$

$P(X = 111) = 3k$

$\Rightarrow \sum_s P(X = s) = 1. \rightarrow 3k + 6k + 3k = 1 \rightarrow k = \frac{1}{12}$

$$\Rightarrow \rho = \begin{array}{c} \left[\begin{array}{cccccccc} 0 & & & & & & & \\ & \frac{1}{12} & & & & & & \\ & & \frac{1}{12} & & & & & \\ & & & \frac{1}{6} & & & & \\ & & & & \frac{1}{12} & & & \\ & & & & & \frac{1}{6} & & \\ & & & & & & \frac{1}{6} & \\ & & & & & & & \frac{1}{4} \end{array}\right] \begin{array}{c} 000 \\ 001 \\ 010 \\ 011 \\ 100 \\ 101 \\ 110 \\ 111 \end{array} \\ \begin{array}{cccccccc} 000 & 001 & 010 & 011 & 100 & 101 & 110 & 111 \end{array} \end{array}$$

$\rho_{ij} = 0 \quad \forall i \neq j$

(b) $\Rightarrow$ Alphabet $= \mathcal{X} = \{00, 01, 10, 11\}$

$\Rightarrow P(X = 00) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$

$P(X = 01) = P(X = 10) = \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}$

$P(X = 11) = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$

$$\Rightarrow \rho = \begin{array}{c|c} \begin{bmatrix} \frac{1}{9} & & & \\ & \frac{2}{9} & & \\ & & \frac{2}{9} & \\ & & & \frac{4}{9} \end{bmatrix} & \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \\ \hline \begin{matrix} 00 & 01 & 10 & 11 \end{matrix} & \end{array}$$

$\rho_{ij} = 0 \quad \forall i \neq j$

(c) $\Rightarrow$ Alphabet $= \mathcal{X} = \{0, 1\}$

$\Rightarrow P(X=0) = P($XOR of all bits of register in (a) $= 0)$

$[0 \wedge 0 = 0,\ 1 \wedge 1 = 0,\ 0 \wedge 1 = 1,\ 1 \wedge 0 = 1]$

$= 0 + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$

[strings — 000, 011, 101, 110]

$\Rightarrow P(X=1) = 1 - P(X=0) = 1 - \frac{1}{2} = \frac{1}{2}$

[strings — 001, 010, 100, 111]

$$\Rightarrow \rho = \begin{array}{c|c} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} & \begin{matrix} 0 \\ 1 \end{matrix} \\ \hline \begin{matrix} 0 & 1 \end{matrix} & \end{array}$$

(d) $\Rightarrow$ Alphabet $= \mathcal{X} = \{0, 1, 2\}$

$\Rightarrow P(X=0) = P(00) = \frac{1}{9}$

$\Rightarrow P(X=1) = P(01) + P(10) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$

$\Rightarrow P(X=2) = P(11) = \frac{4}{9}$

$$\Rightarrow \rho = \begin{array}{c|c} \begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & \frac{4}{9} & 0 \\ 0 & 0 & \frac{4}{9} \end{bmatrix} & \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} \\ \hline \begin{matrix} 0 & 1 & 2 \end{matrix} & \end{array}$$

11. $\Rightarrow$ Let $\mathcal{E}$ be the map.

$$\mathcal{E}(A) = \sum_j K_j A K_j^{\dagger}$$

K_j are arbitrary operators.

$\Rightarrow$ Given A is a positive operator and need not be diagonal.

$\Rightarrow A^{\dagger} = A$; $\langle \psi | A | \psi \rangle \geq 0 \ \forall \ |\psi\rangle \in \mathbb{C}^n$

$$\Rightarrow (\mathcal{E}(A))^{\dagger} = \left(\sum_j K_j A K_j^{\dagger}\right)^{\dagger} = \sum_j (K_j A K_j^{\dagger})^{\dagger}$$

$$= \sum_j (K_j^{\dagger})^{\dagger} A^{\dagger} (K_j)^{\dagger} = \sum_j K_j A^{\dagger} K_j^{\dagger}$$

$$= \sum_j K_j A K_j^{\dagger} = \mathcal{E}(A)$$

$\Rightarrow \mathcal{E}(A)$ is Hermitian ——— ①

$\Rightarrow$ Take any arbitrary $|\psi\rangle \in \mathbb{C}^n$

$$\Rightarrow \langle \psi | \mathcal{E}(A) | \psi \rangle = \cancel{\sum_{\phi} \langle \psi | \mathcal{E}(A) K_j A K_j^{\dagger} |\psi\rangle}$$

$$= \langle \psi | \left(\sum_j K_j A K_j^{\dagger}\right) |\psi\rangle$$

$$= \sum_j \langle \psi | K_j A K_j^{\dagger} |\psi\rangle$$

$$= \sum_j (K_j^{\dagger} |\psi\rangle)^{\dagger} A (K_j^{\dagger} |\psi\rangle)$$

[$\because$ let $|\phi_j\rangle = K_j^{\dagger} |\psi\rangle$]

$$= \sum_j \langle \phi_j | A | \phi_j \rangle \geq 0$$

[$\because$ We know $\langle \phi_j | A | \phi_j \rangle \geq 0 \;\forall\; |\phi_j\rangle \in \mathbb{C}^n$]

$\Rightarrow \langle \psi | \mathcal{E}(A) | \psi \rangle \geq 0 \;\forall\; |\psi\rangle \in \mathbb{C}^n$ —— ②

$\Rightarrow$ By ① and ②,

$\mathcal{E}(A)$ is a positive operator

END